

Chapter – 4

HELIOS WATCH STORE E-COMMERCE DATABASE

Analyzing Functional Dependencies, Redundancy and Data Anomalies

Introduction

The **Helios Watch Store E-Commerce Database** is designed to manage the various activities involved in an online watch retail system, including customer management, product information, brands, categories, shopping carts, orders, payments, shipments, reviews, and warranties. In the previous weeks, the business requirements were analyzed, and the entities, attributes, keys, and relationships required for the database were identified.

In **chapter 4**, the database design is analyzed by identifying the **functional dependencies** for each entity and examining the presence of **data redundancy** and **data anomalies**. Functional dependency helps determine how the attributes within an entity are related to its primary key, ensuring that each non-key attribute is fully dependent on the appropriate primary key. The database is also examined to identify and eliminate redundant data, thereby reducing unnecessary duplication of information. Furthermore, the design is analyzed for **insertion anomalies, update anomalies, and deletion anomalies**, and appropriate measures are taken to minimize these issues through proper database design. This analysis helps verify that the database is well-organized, maintains data consistency, minimizes redundancy, and supports efficient storage, retrieval, and management of information for the Helios Watch Store E-Commerce system.

Customer Entity

Attribute	Key
Customer_ID	Primary Key
First_Name	Not Null
Last_Name	-
Email	Unique Key
Phone	-
Date_of_Birth	-
Age	Check
Address	Not Null

Functional Dependency

Customer_ID → First_Name, Last_Name, Email, Phone, Date_of_Birth, Age, Address

Redundancy Analysis

Customer information is stored once.

Data Anomalies

Insertion Anomaly: No. Data can be inserted without unnecessary duplication.

Update Anomaly: No. Information is updated in one place due to normalization.

Deletion Anomaly: No. Deleting related records does not remove essential master information.

Brand Entity

Attribute	Key
Brand_ID	Primary Key
Brand_Name	Unique Key
Country	-

Functional Dependency

Brand_ID → Brand_Name, Country

Redundancy Analysis

Brand stored once.

Data Anomalies

Insertion Anomaly: No. Data can be inserted without unnecessary duplication.

Update Anomaly: No. Information is updated in one place due to normalization.

Deletion Anomaly: No. Deleting related records does not remove essential master information.

Category Entity

Attribute	Key
Category_ID	Primary Key
Category_Name	Unique Key
Description	-

Functional Dependency

Category_ID → Category_Name, Description

Redundancy Analysis

Category stored once.

Data Anomalies

Insertion Anomaly: No. Data can be inserted without unnecessary duplication.

Update Anomaly: No. Information is updated in one place due to normalization.

Deletion Anomaly: No. Deleting related records does not remove essential master information.

Product Entity

Attribute	Key
Product_ID	Primary Key
Product_Name	-
Brand_ID	Foreign Key
Category_ID	Foreign Key
Price	-
Colors	-
Strap_Material	-
Dial_Color	-
Water_Resistance	-
Stock	Not Null

Functional Dependency

Product_ID → Product_Name, Brand_ID, Category_ID, Price, Colors, Strap_Material, Dial_Color, Water_Resistance, Stock

Redundancy Analysis

Uses Brand_ID and Category_ID to avoid redundancy.

Data Anomalies

Insertion Anomaly: No. Data can be inserted without unnecessary duplication.

Update Anomaly: No. Information is updated in one place due to normalization.

Deletion Anomaly: No. Deleting related records does not remove essential master information.

Cart Entity

Attribute	Key
Cart_ID	Primary Key
Customer_ID	Foreign Key
Grand_Total	-
Total_Items	-

Functional Dependency

Cart_ID → Customer_ID, Grand_Total, Total_Items

Redundancy Analysis

No redundancy.

Data Anomalies

Insertion Anomaly: No. Data can be inserted without unnecessary duplication.

Update Anomaly: No. Information is updated in one place due to normalization.

Deletion Anomaly: No. Deleting related records does not remove essential master information.

Order Entity

Attribute	Key
Order_ID	Primary Key
Customer_ID	Foreign Key
Order_Date	-
Total	-
Order_Status	-
Total_Amount	-

Functional Dependency

Order_ID → Customer_ID, Order_Date, Total, Order_Status, Total_Amount

Redundancy Analysis

No redundancy.

Data Anomalies

Insertion Anomaly: No. Data can be inserted without unnecessary duplication.

Update Anomaly: No. Information is updated in one place due to normalization.

Deletion Anomaly: No. Deleting related records does not remove essential master information.

Cart_Item Entity

Attribute	Key
Cart_Item_ID	Primary Key
Cart_ID	-
Product_ID	-
Quantity	-

Unit_Price	-
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Functional Dependency

Cart_Item_ID → Cart_ID, Product_ID, Quantity, Unit_Price

Redundancy Analysis

No redundancy.

Data Anomalies

Insertion Anomaly: No. Data can be inserted without unnecessary duplication.

Update Anomaly: No. Information is updated in one place due to normalization.

Deletion Anomaly: No. Deleting related records does not remove essential master information.

Order_Item Entity

Attribute	Key
Order_Item_ID	Primary Key
Order_ID	Foreign Key
Product_ID	Foreign Key
Quantity	-
Unit_Price	-

Redundancy Analysis

No redundancy.

Data Anomalies

Insertion Anomaly: No. Data can be inserted without unnecessary duplication.

Update Anomaly: No. Information is updated in one place due to normalization.

Deletion Anomaly: No. Deleting related records does not remove essential master information.

Payment Entity

Attribute	Key
Payment_ID	Primary Key
Order_ID	Foreign Key
Payment_Method	-
Payment_Status	-
Payment_Date	-

Amount	-
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Functional Dependency

Payment_ID → Order_ID, Payment_Method, Payment_Status, Payment_Date, Amount

Redundancy Analysis

No redundancy.

Data Anomalies

Insertion Anomaly: No. Data can be inserted without unnecessary duplication.

Update Anomaly: No. Information is updated in one place due to normalization.

Deletion Anomaly: No. Deleting related records does not remove essential master information.

Review Entity

Attribute	Key
Review_ID	Primary Key
Customer_ID	Foreign Key
Product_ID	Foreign Key
Rating	-
Review_Text	-
Review_Date	-

Functional Dependency

Review_ID → Customer_ID, Product_ID, Rating, Review_Text, Review_Date

Redundancy Analysis

No redundancy.

Data Anomalies

Insertion Anomaly: No. Data can be inserted without unnecessary duplication.

Update Anomaly: No. Information is updated in one place due to normalization.

Deletion Anomaly: No. Deleting related records does not remove essential master information.

Shipment Entity

Attribute	Key
Shipment_ID	Primary Key
Order_ID	Foreign Key

Delivery_Address	-
Shipment_Date	-
Delivery_Date	-
Shipment_Status	-

Functional Dependency

Shipment_ID → Order_ID, Delivery_Address, Shipment_Date, Delivery_Date, Shipment_Status

Redundancy Analysis

No redundancy.

Data Anomalies

Insertion Anomaly: No. Data can be inserted without unnecessary duplication.

Update Anomaly: No. Information is updated in one place due to normalization.

Deletion Anomaly: No. Deleting related records does not remove essential master information.

Warranty Entity

Attribute	Key
Warranty_ID	Primary Key
Product_ID	Foreign Key
Warranty_Period	-
Start_Date	-
End_Date	-

Functional Dependency

Warranty_ID → Product_ID, Warranty_Period, Start_Date, End_Date

Redundancy Analysis

No redundancy.

Data Anomalies

Insertion Anomaly: No. Data can be inserted without unnecessary duplication.

Update Anomaly: No. Information is updated in one place due to normalization.

Deletion Anomaly: No. Deleting related records does not remove essential master information.

Conclusion

The **Helios Watch Store E-Commerce Database** was successfully analyzed by identifying the functional dependencies, examining data redundancy, and evaluating possible insertion, update, and deletion anomalies for each of the twelve entities. The analysis confirmed that every entity has a well-defined primary key, and all non-key attributes are functionally dependent on their respective primary keys. The use of foreign keys establishes appropriate relationships between entities and minimizes unnecessary duplication of data.

The database design effectively reduces redundancy by storing information only once and referencing related data through primary and foreign keys. This approach helps prevent insertion, update, and deletion anomalies, thereby improving data consistency, integrity, and ease of maintenance. Overall, the database is well-structured and suitable for efficiently managing the operations of the Helios Watch Store E-Commerce system.